

Daksh K. Shah

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EDUCATION

New York University, Courant Institute

New York, NY

Master of Science in Computer Science (Incoming Fall 2026)

May 2028

University of California, Santa Cruz

Santa Cruz, CA

B.S. in Computer Science; B.S. Math Theory and Computation — GPA: 3.89

June 2026

Honors/Awards: Chancellor's Award for Undergraduate Research, Dean's List (8x)

Thesis: *K-Material SDFs for Neural Attenuation Fields*

Relevant Coursework: Computer Graphics, Deep Learning, Numerical Methods, Advanced Linear Algebra, Distributed Systems, Advanced Computer Vision (Grad), Statistical Machine Learning (Grad), Artificial Intelligence (Grad), Advanced ML: NeuroSymbolic AI & Differentiable Programming (Grad), Applied Bayesian Statistics (Grad)

SKILLS

ML Frameworks and Compute: PyTorch, CUDA, Huggingface Transformers, LoRA, WandB

ML & 3D Computer Vision: 3D Gaussian Splatting, NeRFs, Neural SDFs, Vision-Language Models, Self-supervised Learning, Representation Learning

MLOps and Infrastructure: Kubernetes Docker, Kubeflow, DDP, Distributed Systems

Graphics and Compute Libraries: OpenCV, WebGL, OpenGL, Three.js

PUBLICATIONS

K-NeAS: Scalable Multi-Material CT Reconstruction using Neural SDFs — *Daksh K. Shah, Emmanouil Nikolakakis, Razvan V. Marinescu* — Under Review, Off-Grid Workshop, MICCAI (2026)

ReNAF: Regularized Neural Attenuation Fields with 3D Reconstruction Priors for Sparse-View CT — *Emmanouil Nikolakakis, Daksh K. Shah, Julia Wong, Julie Digne, Razvan V. Marinescu* — In Preparation

Video Super-Resolution Benchmark: Evaluating Spatial Fidelity and Temporal Coherence Tradeoffs — *Daksh K. Shah, Dylan J. Louie* — Technical Report, UC Santa Cruz (2025)

RESEARCH EXPERIENCE

Research Assistant (PI: Razvan Marinescu)

January 2024 – Present

BiomedAI Lab, UC Santa Cruz

Santa Cruz, CA

- Containerized the ReNAF training pipeline in Docker and wrote parallelized Kubernetes job scripts to reserve a dedicated 8-GPU node on the NRP Nautilus cluster, enabling overnight unattended validation and supplementing a PhD student's manual runs on shared lab hardware; reran and validated metrics across 200+ experiments.

Research Assistant (PI: James Davis)

April 2025 – June 2026

Visualization and Interactive Systems (VIS) Lab, UC Santa Cruz

Santa Cruz, CA

- Benchmarked 2-3 sparse-view 3D Gaussian Splatting pipelines, including original 3DGS and Meta's official Aria splatting pipeline, across 1-2 egocentric scenes to evaluate reconstruction quality from Project Aria sensor data.
- Advanced to an advisory role leading a weekly reading group of 5 students, covering 32 frontier papers over 6 months on Embodied Navigation, Vision-Language Models, and 3D Reconstruction.

PROJECTS

K-NeAS: Scalable Multi-Material CT Reconstruction

PyTorch, CUDA, Kubernetes

Nov. 2025 – July 2026

- Engineered a unified multi-head implicit neural representation with a shared latent backbone and K output heads, improving 3D PSNR by up to 1.8 dB over baseline NeAS as material count scales from 1 to 4, and by 1.1 dB at 10-view sparse acquisitions on clinical CBCT data.
- Designed a differentiable K-material soft selector enabling reconstruction across an arbitrary number of tissue types, where baseline methods are hard-capped at 2 materials.
- Automated attenuation bound estimation via unsupervised GMM clustering, eliminating per-scene manual hyperparameter tuning required by the baseline.

MaskSDF: Physics-Constrained 3D Priors for 2D-to-3D Generation

April 2026 – Present

- Designing a feed-forward Point-ViT masked autoencoder with EG3D-style triplanes to reconstruct watertight surfaces from heavily masked (80%+) point clouds, enforcing geometry via Eikonal and normal regularization.
- Diagnosed a Ceph I/O bottleneck on small files and sharded 100K files to Hugging Face datasets, leveraging faster sequential reads and DDP to reduce training cost from 400 to 160 GPU-hours.

Distributed Sharded Key-Value Store | Go, Docker

Jan. 2025 – Mar. 2025

- Built a fault-tolerant, causally-consistent sharded key-value store in Go with vector-clock metadata, consistent hashing, and gossip-based anti-entropy; tested across 100+ containers, 12 shards, with live rebalancing.

PROFESSIONAL EXPERIENCE

Software R&D Intern

Jan. 2026 – June 2026

Keysight Technologies

Remote

Prototyped a cross-platform, standalone port of Keysight's Live Results Viewer plugin using Avalonia and NATS in C#, achieving full feature parity with the desktop version and establishing an internal research pathway for converting Keysight's legacy desktop plugins to the web.

LEADERSHIP, TEACHING, & SERVICE

Teaching Assistant – Baskin School of Engineering, UC Santa Cruz

April 2023 – June 2026

- Supported 3000+ students across 7 courses and 10 terms, including Computer Vision and Deep Learning.
- Built automated Piazza mirroring and CSE 160 resource sites; earned Best Tutor for three consecutive terms.

Instructional Officer – Santa Cruz Artificial Intelligence

Aug. 2025 – June 2026

- Delivered technical workshops on neural network architectures and training workflows to 50+ participants.
- Led a mini-hackathon and workshop on MLOps methodologies covering Kubernetes, Docker, and LoRA.